ICLR 2027 submission 56173: the reviews, our replies and the revision

Each point quotes the reviewer (in red, from Reviews.pdf), then gives the reply we plan to send, then says what changed in the revised paper (in green). Figure and table numbers refer to the revised paper. After this document come twelve figures, one per group of claims, each showing the reviewer's words, the submitted passage and the revised passage side by side.

Reviewer k4nd

W1: the residual does not establish a distinct representation of plurality

> *"The proposed residual does not establish a distinct representation of plurality. … The paper does not establish that it specifically measures whether the named concepts appear as distinct objects."*

We agree that r_t in Equation 9 is the entire difference between the joint-prompt and product-of-experts predictions, and that adding it in full recovers the joint prompt by construction. We do not claim that r_t represents plurality alone. A product of experts drops the interaction the joint prompt carries, and that interaction can hold several kinds of meaning: a relation between the concepts ("a dog on a cat"), an effect of one on the other ("a camel in the rain" should be wet), and the layout and style of the scene.

Our study concerns one part of this interaction: plurality, whether two named concepts appear as two objects rather than one hybrid. The training pairs were chosen for it, as pairs the product fuses into a hybrid.

Within that scope, the evidence is that the correction is specific and early, not generic. A pair's own r_t restores the second object, while another pair's, another seed's, or a time-reordered one does not (Figure 2). The first ten of 50 steps are enough (Figures 3 and 4). At partial strength (λ of 0.6 to 0.65 in Figure 2) the second object appears while the layout and style still look like the product's render.

We agree this does not separate plurality from everything else r_t carries. Section 3 now states this scope explicitly, and the contributions are rewritten to claim only what the experiments show.

What changed in the revision. Section 3 now says r_t is the whole interaction the product drops and scopes the paper to plurality; the contributions claim only what the experiments show.

W2: methodological novelty

> *"The methodological novelty is insufficiently established."*

The method is deliberately simple: a standard adapter trained by prediction matching. Its contribution is the target. Existing corrections of product-of-experts sampling keep the product as the target and

change how it is sampled (Du et al., 2023; Skreta et al., 2024, 2025; Dutta et al., 2026). We instead adapt the experts so that their product reproduces the joint prompt's prediction, and the joint prompt is never evaluated at inference. With this simple objective, plurality emerges on concept pairs the adapter never trained on (Section 4.1, Figure 5). Section 2 now states this distinction.

We agree that fusion under composition was identified before (Bradley et al., 2025), and we cite it as such. What we add is a measurement of the missing term during sampling: it is specific to the pair and seed, and concentrated in the first ten of 50 steps (Figures 2 to 4).

Table 1 compares the adapter with PoE and CO3 on the same seeds, and Figure 5 adds a Langevin corrector. The ablation the reviewer describes, the same joint-prompt supervision on pairs the product does not fuse, would separate the effect of the supervision from that of the plurality pairs; we have not run it and name it as future work.

What changed in the revision. Section 2 now says prior corrections keep the product as the target while this paper changes the target. The ablation separating joint-prompt supervision from the choice of pairs is left as future work.

W3: no systematic quantitative evaluation of the adapter

> *"The learned adapter lacks systematic quantitative evaluation."*

We agree, and the revision adds this evaluation. The adapter, the seeds and the success bar were all fixed before any held-out score was read (Section 4.1).

Table 1 scores every method on the 276 Attend-and-Excite prompts (66 animal pairs, 144 animal-object pairs, 66 object pairs) at seeds 9 to 12, with composition measured by BLIP-VQA and image quality by CLIP score and ImageReward.

The training and held-out pairs, with every seed, are released with the code.

Hybrids, missing concepts and additional objects are counted per setting in Figure 11 (Appendix C.2).

What changed in the revision. Table 1 scores every method on the 276 Attend-and-Excite prompts with BLIP-VQA, CLIP score and ImageReward. Figures 11 and 12 score 53 held-out pairs as the adapter acts on fewer steps. Section 4.1 opens with the evaluation protocol.

W4: method details poorly presented and sometimes inconsistent

> *"Section 4 describes target matching and provides the teacher prediction and sampling rule, but it does not explicitly formulate the training loss. The squared-error objective appears in the training diagram in Appendix A, Figure 7." "The main text describes individual adapted network evaluations as returning the correction, whereas Appendix B defines them as adapted denoising predictions whose composition induces the correction. Section 4 refers to 'Figure 10,' although the actual training diagram is Figure 7." "The repetitive claims about learning an abstract concept occupy space that could instead clarify the objective, training data, evaluation protocol, and limitations."*

The training loss is not stated in Section 4

Section 4 now states the training loss as Equation 12: the squared error between the adapted composition and the joint-prompt prediction, over the cached states of every training pair and seed at the first ten of the 50 sampling steps. The training diagram is removed, since Equation 12 now states the objective in the main text.

Evaluations described as returning the correction

Section 4 now says each adapted evaluation returns a noise prediction, as Appendix B derives. At inference we run the usual product-of-experts step on c_1, c_2 and the null prompt with the adapter switched on: three predictions per step, and never the joint prompt. The joint prompt's prediction is the product's prediction plus the correction r_t, so matching it trains the adapter to add r_t. At inference, switching the adapter on adds it. The reference that printed as "Figure 10" is removed.

Repetitive claims about learning an abstract concept

We removed the repeated claims about learning an abstract concept from Sections 3 and 4 and the Discussion. The introduction now states what the correction is and what it restores (see W1). Section 4 now describes the training data and how pairs were chosen (Figure 7). Appendix C.4 (Figure 16) and a new Discussion paragraph show the adapter on three concepts, including where it fails: it separates the concepts in every case but stops at two animals. Section 4.1 now opens with the evaluation protocol: one adapter chosen by a fixed rule, the same seed for every method, the prompt sets and held-out pairs with their seeds, the scorers, and the renders behind Table 1 in the appendix.

What changed in the revision. Section 4 states the training loss as Equation 12 and says each adapted evaluation returns a noise prediction; inference is the usual product-of-experts step with the adapter switched on. The training diagram and the "Figure 10" reference are gone, and the repeated abstract-concept claims are cut.

W5: the qualitative baseline comparison and its starting noise

> *"The Figure 5 caption states that CO3 uses different starting noise, while Section 4.1 states that all examples use the same initial seeds."*

Every method now uses the same seed for each prompt, with every column of a row in Figure 5 starting from the same latent, and Section 4.1 says so. CO3's column is marked † because its later resampling noise comes from the seed, as in its released code. Table 1 makes the comparison over 4 seeds, and Figures 11 and 12 over 8, with the images behind them in the appendix.

What changed in the revision. Figure 5's caption and Section 4.1 say every column starts from the same latent; CO3 is marked with a dagger for its later resampling noise.

Q1: evidence that the residual measures plurality

> *"What evidence establishes that the residual in Equation (9) specifically measures plurality rather than the broader difference between joint and separate conditioning?"*

Answered under W1 above.

What changed in the revision. As for W1.

Q2: aggregate results over held-out concepts and seeds

> *"Can the authors provide aggregate results for the learned adapter and baselines over documented held-out concepts and multiple seeds, including identity errors, additional objects, and style-dependent failures?"*

Answered under k4nd W3: Table 1, and Figure 11, which counts the failure types the reviewer names.

What changed in the revision. Table 1, and Figures 11 and 12 on the held-out pairs.

Q3: pairs retained when only CO3 succeeds

> *"Training pairs are retained when either joint prompting or CO3 succeeds, but supervision comes from the frozen joint-prompt prediction. How are pairs handled when only CO3 succeeds?"*

We have addressed this in Section 4 and Figure 7. Every training target is the frozen model's joint-prompt prediction; no CO3 picture is a target. CO3 was a second view during screening: a pair entered the training set when the joint prompt drew both named things as separate objects. Figure 7 shows the joint prompt beside CO3 for five candidate pairs, including crow and raven, dropped because the two birds are hard to tell apart. Where CO3 drew a kept pair more clearly on a seed (seed 3 in Figure 7), that seed's target is still the joint prompt's picture.

What changed in the revision. Section 4 describes the training set; Figure 7 shows the joint prompt beside CO3 during screening. No CO3 picture is a training target.

Q4: the objective in the main text, and the inconsistencies

> *"Can the authors state the optimization objective explicitly in the main text and resolve the inconsistencies concerning network outputs, figure references, and baseline starting noise?"*

The objective is Equation 12 in Section 4. The network outputs and the figure reference are resolved as described under W4. Baseline starting noise is resolved as described under W5.

What changed in the revision. As for W4 and W5.

Reviewer UHz6

W1: the adapter is evaluated only qualitatively

> *"The trained adapter, which serves as the main contribution of the paper is only evaluated qualitatively. … In addition to the examples being limited to similar entities (cat, dog), there are no standardized benchmark evaluations such as T2I-CompBench or GenEval counting."*

The revision evaluates the adapter quantitatively on the 276 prompts of the Attend-and-Excite benchmark, which cover animal pairs, animal-object pairs and object pairs (Table 1), and on 53 held-out pairs as the adapter acts on fewer steps (Figures 11 and 12). Table 2 reports GenEval's two-object, attribute-binding and position tasks and T2I-CompBench's colour and shape sets, scored by their official evaluators. Appendix C.5 shows three concepts.

What changed in the revision. Table 1 adds three scorers on 276 prompts; Table 2 adds GenEval and T2I-CompBench.

W3: comparisons with samplers and attention-based methods, and scope

> *"There are no quantitative comparisons with the compositional samplers discussed in Sec. 2.2, and CO3 appears only as one qualitative column. Attention-based methods that address missing or merged objects under joint prompts (e.g., Attend-and-Excite, CONFORM) target the same failure mode and are not discussed. The study is also limited to SDXL and to pairs of two concepts, mostly animals."*

Table 1 now compares quantitatively with CO3 on the same seeds, and Figure 5 adds a Langevin corrector. Attend-and-Excite and CONFORM are now discussed in Section 2: they adjust cross-attention under a single joint prompt, while our setting has no joint prompt at inference. Beyond pairs of animals, Table 1 covers objects and Appendix C.4 three concepts.

What changed in the revision. Figure 5 adds ToMe and a Langevin corrector; Table 1 compares with CO3 and ToMe; Section 2 discusses Attend-and-Excite and CONFORM; Appendix C.4 shows three concepts.

The training filter

> *"In addition, the training filter keeps a pair if either the joint prompt or CO3 succeeds, but the target is always the joint prompt (Eq. 11). Pairs that pass only through CO3 would therefore supervise toward a target without plurality."*

We have addressed this in Section 4 and Figure 7. Every training target is the frozen model's joint-prompt prediction; no CO3 picture is a target. CO3 was a second view during screening: a pair entered the training set when the joint prompt drew both named things as separate objects. Figure 7 shows the joint prompt beside CO3 for five candidate pairs, including crow and raven, dropped because the two birds are hard to tell apart. Where CO3 drew a kept pair more clearly on a seed (seed 3 in Figure 7), that seed's target is still the joint prompt's picture.

What changed in the revision. As for k4nd Q3.

Presentation issues

> *"There are presentation issues: Sec. 4 refers to a Figure 10 that does not exist, and several sentences are broken (e.g., "inside create hybrid objects" in Sec. 5)."*

The reference that printed as "Figure 10" in Section 4 is removed, and the Discussion now reads "and instead create hybrid objects".

What changed in the revision. The "Figure 10" reference is removed, and the Discussion reads "and instead create hybrid objects".

Q1: the 48 training pairs, the held-out pairs, and whether cat and dog is trained on

> *"Could the authors list the 48 training pairs and the held-out pairs, and clarify whether cat/dog is in the training set?"*

The training set is defined by a criterion, not by the particular pairs. A pair is kept when the frozen model's joint prompt draws both named concepts as separate objects, and dropped when the two

concepts cannot be told apart (Section 4, Figure 7). The same criterion applies whether it yields 48 pairs or a thousand, and 48 is where we stopped collecting. The method relies on the criterion, not on any specific pair.

Cat and dog is not in the training set. It is held out, and no training pair contains a cat or a dog. The held-out pairs are released with the code, and no pair in Figure 5 was trained on.

For reproducibility, the 48 training pairs are: (two animals, 28) badger and magpie, bear and boar, bear and heron, beaver and raccoon, beaver and woodpecker, bison and weasel, buffalo and hippo, bull and rooster, camel and pelican, chicken and rabbit, deer and heron, deer and wolf, donkey and duck, donkey and rooster, emu and koala, giraffe and hyena, giraffe and ostrich, giraffe and zebra, gorilla and chimpanzee, hedgehog and robin, horse and pig, horse and sheep, impala and jackal, langur and peacock, lion and meerkat, lion and oryx, moose and chipmunk, peacock and pig; (an animal and an object, 3) hamster and carrot, pony and red barn, snail and lantern; (two objects, 17) accordion and scarecrow, bicycle and birdhouse, birdcage and hay bale, cactus and cooler box, coat rack and potted plant, guitar and traffic cone, hay bale and ukulele, lighthouse and sailboat, mailbox and pumpkin, microscope and test tube, rocking chair and cactus, soccer ball and park bench, suitcase and globe, teddy bear and birthday cake, telescope and beach ball, typewriter and cactus, windmill and tractor. They are also released with the code as the configuration file the trainer reads.

What changed in the revision. Section 4 describes the training set by kind; the 48 pairs are listed in the reply and released with the code.

Q2: success rate against PoE and CO3 over all held-out pairs and seeds

> *"What is the success rate of the adapter compared with PoE and CO3 over all held-out pairs and multiple seeds?"*

Answered under k4nd W3: Table 1, and Figures 11 and 12 for the held-out pairs at eight seeds.

What changed in the revision. Table 1, and Figures 11 and 12 on the held-out pairs.

Q3: restricting the adapter to the early steps

> *"Given the finding in Sec. 3, does restricting the adapter to the early steps change the failure modes in Fig. 7?"*

We tested this directly. Figure 11 counts each failure type on the 53 held-out pairs at seeds 9 to 16 with the adapter switched on for the first 10, the first 25 or all 50 of the 50 steps, and Figure 13 shows the images. The first 25 steps give the same result as all 50. With the first 10 only, the detector finds two animals far less often (Figure 12a). The cow and horse row of Figure 13 shows that the species substitution of the submitted Fig. 7 (Figure 15 in the revision) persists at every window, so it does not come from the late steps.

What changed in the revision. Figures 11, 12 and 13 show the adapter switched on for the first 10, the first 25 or all 50 steps.

Q4: pairs that do not compete for the same region

> *"Does the adapter degrade PoE on pairs that do not compete for the same region, such as foreground/background pairs like lighthouse and sea?"*

No. Among our held-out pairs, 21 have concepts that do not contend for one region, including lighthouse and a stormy sea. On these pairs the adapter scores the same as the plain product within the 95% intervals (BLIP-VQA, seeds 9 to 16), so on pairs the product already composes it neither helps nor harms.

What changed in the revision. No change to the paper; the reply answers from the held-out pairs.

Reviewer PrzE

W1: lack of quantitative benchmarks

> *"The paper does not report a single quantitative metric. Standard benchmarks such as T2I-Compench, GenEval and standard metrics such as CLIP, Imagereward are completely absent in this paper. The empirical results and a few qualitative results are insufficient for the claim in this paper."*

The revision adds quantitative evaluation. Table 1 scores every method on the 276 Attend-and-Excite prompts with BLIP-VQA (the T2I-CompBench composition metric), CLIP score and ImageReward. Table 2 reports GenEval's two-object, attribute-binding and position tasks and T2I-CompBench's colour and shape sets, scored by their official evaluators.

What changed in the revision. Table 1 adds BLIP-VQA, CLIP score and ImageReward on 276 prompts; Table 2 adds GenEval and T2I-CompBench.

W2: narrow baselines and missing details

> *"There is only CO3 baseline compared, the paper doesn't provide other corrected methods such as MCMC corrector. The paper only compares the baseline CO3 without the corrector they mention. The 48 concept pairs are not listed so it's impossible to see what the adapter is trained on."*

Figure 5 now includes a product of experts with a Langevin corrector (Du et al., 2023), the MCMC correction the paper discusses, run at its published setting from the same starting latent, beside CO3, ToMe and our adapter. Table 1 compares with CO3 on every image. Table 1 adds ToMe, run on SDXL from the same starting latents. Section 4 now describes the training set and how pairs were chosen (Figure 7); the 48 pairs are listed in our reply to Reviewer UHz6 and released with the code.

What changed in the revision. Figure 5 adds ToMe and a Langevin corrector; Table 1 adds ToMe; Section 4 describes the training set.

Q1: quantitative benchmark

> *"What are the evaluation scores for the proposed method? Can you show the results on some benchmarks and compare your method with other standard baselines?"*

Answered under W1 and W2: Tables 1 and 2.

What changed in the revision. As for W1 and W2.

Q2: multiple concepts

> *"How does your method perform when composing more than two concepts? Can you show some results for more than two concepts?"*

Appendix C.4 (Figure 16) shows one triple of each kind. The plain product fuses the three concepts or drops one; the adapter, trained only on pairs, separates them in every case and draws all three named things when the triple includes objects, but with two or three animals it draws only two. The Discussion now addresses this.

What changed in the revision. Appendix C.4 and a Discussion paragraph show the adapter on three concepts.

G1: What the residual measures

k4nd weakness 1 and question 1 · status: answered

THE REVIEWER'S POINT

k4nd, weakness 1

1. The proposed residual does not establish a distinct representation of plurality.

Definition 1.1 describes a semantic property of the generated image. However, Equation (9) defines the "plurality term" as the entire difference between joint-prompt and PoE guided predictions. This residual can contain differences in object identity, appearance, layout, style, and other interactions between the prompts. The paper does not establish that it specifically measures whether the named concepts appear as distinct objects.

Moreover, substituting Equation (9) into Equation (10) gives

$$(1-\lambda)\hat{\epsilon}_{\text{PoE}} + \lambda\hat{\epsilon}_{\text{Mono}}$$

Consequently, the full correction recovers the joint-prompt prediction by construction, as the authors acknowledge. The partial-strength and early-step experiments provide empirical observations, but they do not resolve whether the residual isolates plurality. The stronger claims about measuring and learning an abstract concept therefore exceed the evidence.

Reviews.pdf, page 4 (no text layer; passage located by OCR)

K-W1a

K-W1b

K-W1b

k4nd, question 1

- What evidence establishes that the residual in Equation (9) specifically measures plurality rather than the broader difference between joint and separate conditioning?

Reviews.pdf, page 5 (no text layer; passage located by OCR)

K-Q1

this claim

by construction

THE SUBMITTED WEAKNESS

Introduction, the contributions

as "plurality" and define this concept for the context of compositional diffusion in Definition 1.1.

Definition 1.1: Plurality

The property of a prompt and image pair where each named concept in the prompt is rendered as a distinct object in the generated scene.

We show that plurality can be learned by an adapter network layer with a finite set of examples from conflicting concept pairs. Further, the learned concept of plurality within the latent space generalises to novel concept pairs. Thus, our method is able to induce plurality zero-shot for concept combinations that cannot be found in the training data and that joint prompting is unlikely to generalise to. We make three primary contributions:

- We measure the abstract concept of plurality which is already present in the latent space of a pretrained LDM such as Stable Diffusion XL.

56173_Restoring_plurality_in_c.pdf (submitted), page 2

measures plurality itself

Section 3, the plurality term

We define the plurality term r_t as the difference between the model's guided prediction under the joint prompt and its guided product-of-experts prediction. Denoting $\hat{\epsilon}_{\text{Mono}}$ as the joint-prompt prediction and evaluating both model predictions at the same latent representation x_t gives:

$$r_t = \hat{\epsilon}_{\text{Mono}}(x_t, t) - \hat{\epsilon}_{\text{PoE}}(x_t, t). \quad (9)$$

Measuring r_t requires a model which is able to produce a reasonable image conditioned on the joint prompt. This is likely to only be true for common concept combinations and places the burden on data and training, rather than inference, which is counter to the purpose of compositional diffusion. To begin our exploration, we add the measured plurality term to the product-of-experts and control the strength of the correction with λ:

56173_Restoring_plurality_in_c.pdf (submitted), page 4

whole difference, named plurality

rewritten

THE REVISED ANSWER

now named for what it is

Section 3, what the term is

We define the plurality term r_t as the difference between the model's guided prediction under the joint prompt and its guided product-of-experts prediction. Denoting $\hat{\epsilon}_{\text{Mono}}$ as the joint-prompt prediction and evaluating both model predictions at the same latent representation x_t gives:

$$r_t = \hat{\epsilon}_{\text{Mono}}(x_t, t) - \hat{\epsilon}_{\text{PoE}}(x_t, t). \quad (9)$$

The term r_t is the whole interaction that the product drops; everything the joint prompt implies beyond its separate concepts. That can include a relation between the concepts (a dog on a cat), an effect of one on the other (a camel in the rain is wet), and the layout and style of the scene. This paper studies one part of it, plurality: whether two named concepts appear as two objects rather than one hybrid. The training pairs are pairs the product

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 5

the whole interaction

scoped to plurality

Introduction, the contributions

- We measure, at every denoising step, the difference between SDXL's joint-prompt prediction and its product-of-experts prediction, and show that a pair's own correction over the first ten of 50 steps restores plurality.
- We train an adapter so that the product of the separate prompts reproduces the joint-prompt prediction, removing the joint prompt at inference.
- We show that the adapter restores plurality on concept pairs it never trained on.

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 3

claims only what is measured

The reply we plan to send (replies/k4nd-and-uhz6.md: k4nd W1 and Q1)

answered

W1: the residual does not establish a distinct representation of plurality

We agree that r_t in Equation 9 is the entire difference between the joint-prompt and product-of-experts predictions, and that adding it in full recovers the joint prompt by construction. We do not claim that r_t represents plurality alone. A product of experts drops the interaction the joint prompt carries, and that interaction can hold several kinds of meaning: a relation between the concepts ("a dog on a cat"), an effect of one on the other ("a camel in the rain" should be wet), and the layout and style of the scene.

Our study concerns one part of this interaction: plurality, whether two named concepts appear as two objects rather than one hybrid. The training pairs were chosen for it, as pairs the product fuses into a hybrid.

Within that scope, the evidence is that the correction is specific and early, not generic. A pair's own r_t restores the second object, while another pair's, another seed's, or a time-reordered one does not (Figure 2). The first ten of 50 steps are enough (Figures 3 and 4). At partial strength (λ of 0.6 to 0.65 in Figure 2) the second object appears while the layout and style still look like the product's render.

We agree this does not separate plurality from everything else r_t carries. Section 3 now states this scope explicitly, and the contributions are rewritten to claim only what the experiments show.

Q1: evidence that the residual measures plurality

Answered under W1 above.

Changes behind this page: Section 3 says r_t is the whole interaction the product drops and scopes the paper to plurality; the contributions claim only what the experiments show.
Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.

G2: What is new in the method

k4nd weakness 2 · status: partly answered (the supervision ablation is future work)

THE REVIEWER'S POINT

k4nd, weakness 2

2. The methodological novelty is insufficiently established.

The implemented method matches a LoRA-adapted PoE prediction to a frozen joint-prompt teacher on cached latent states. The paper does not clearly establish what methodological contribution this provides beyond supervised prediction matching with a standard adapter.

The central observation that compositional sampling can fuse competing objects is already attributed to prior work in the manuscript. Introducing a new name for this failure and its correction does not, by itself, establish a substantive contribution. Stronger comparisons and ablations are needed to identify which benefits arise from joint-prompt supervision and which require the proposed plurality formulation.

Reviews.pdf, page 4 (no text layer; passage located by OCR)

K-W2

K-W2

the novelty claimed

THE SUBMITTED WEAKNESS

Introduction, the contributions

- We show that adding a plurality term to PoE restores composition and that this plurality term is learnable by an adapter network layer, alleviating the need for joint prompting at inference time
- We show that the learned concept of plurality can be generalised to novel concept pairs at inference time, providing PoE with the capacity to generate coherent, multi-object scenes through compositional generalisation of individual objects and the abstract notion

56173_Restoring_plurality_in_c.pdf (submitted), page 3

framed as a new concept

stated as

not run

THE REVISED ANSWER

Section 2, prior corrections

what is new: the target

et al., 2025b). These methods keep the product as the target and change how it is sampled; we change the target instead, adapting the experts so that their product reproduces the joint prompt while inference still evaluates only the separate prompts. With this simple objective, plurality emerges on concept pairs the adapter never trained on (Section 4.1).

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 4

Not in the revision

The ablation k4nd asks for (the same joint-prompt supervision on pairs the product does not fuse) was not run. The reply names it as future work.

no crop: this panel states what the claim table records

The reply we plan to send (replies/k4nd-and-uhz6.md: k4nd W2)

partly answered

W2: methodological novelty

The method is deliberately simple: a standard adapter trained by prediction matching. Its contribution is the target. Existing corrections of product-of-experts sampling keep the product as the target and change how it is sampled (Du et al., 2023; Skreta et al., 2024, 2025; Dutta et al., 2026). We instead adapt the experts so that their product reproduces the joint prompt's prediction, and the joint prompt is never evaluated at inference. With this simple objective, plurality emerges on concept pairs the adapter never trained on (Section 4.1, Figure 5). Section 2 now states this distinction.

We agree that fusion under composition was identified before (Bradley et al., 2025), and we cite it as such. What we add is a measurement of the missing term during sampling: it is specific to the pair and seed, and concentrated in the first ten of 50 steps (Figures 2 to 4).

Table 1 compares the adapter with PoE and CO3 on the same seeds, and Figure 5 adds a Langevin corrector. The ablation the reviewer describes, the same joint-prompt supervision on pairs the product does not fuse, would separate the effect of the supervision from that of the plurality pairs; we have not run it and name it as future work.

Changes behind this page: Section 2 says prior corrections keep the product as the target while this paper changes the target.

Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.

G3: How often the adapter works on unseen pairs

k4nd weakness 3 and question 2, UHz6 weakness 1 and question 2 · status: answered

THE REVIEWER'S POINT

k4nd, weakness 3

3. The learned adapter lacks systematic quantitative evaluation.

The principal results in Section 4.1 are qualitative. Figure 5 presents four concept pairs, while Figures 6 and 7 provide selected ablations and failure examples. There are no aggregate benchmark results establishing how frequently the learned adapter improves composition relative to joint prompting, PoE, or CO3. K-W3a

The paper does include detector-based counts, manually verified, in Figure 4. However, these concern an oracle correction derived from the joint prompt for one cat–dog pair and four seeds. They do not establish the effectiveness or generalization of the learned adapter. K-W3b

This omission is particularly serious given the reported substitutions, additional objects, and style-dependent failures. Such examples do not themselves disprove generalization, but their prevalence must be measured. The paper needs aggregate success and failure rates across a sufficiently large, documented test set and multiple seeds, together with uncertainty estimates and image-quality evaluation. K-W3c

Reviews.pdf, page 4 (no text layer; passage located by OCR)

k4nd, weakness 3 (continued)

The manuscript states that evaluation concepts and starting noise are excluded from adapter training. However, the complete splits and aggregate results needed to assess this claim are missing. A few successful held-out examples cannot substantiate the broad claim that the adapter learns a general, transferable concept of plurality. Likewise, one failed joint-prompt image does not establish that a concept combination is poorly represented in pretraining or generally beyond joint prompting. K-W3d

Reviews.pdf, page 5 (no text layer; passage located by OCR)

UHz6, weakness 1 and question 2

- The trained adapter, which serves as the main contribution of the paper, is only evaluated qualitatively. In Fig. 5 and 6, the authors provide examples from the trained adapter but the results are not enough to claim the effectiveness of the method as it is difficult to evaluate its generalization. In addition to the examples being limited to similar entities (cat, U-W1a

Reviews.pdf, page 3 (no text layer; passage located by OCR)

UHz6, question 2

- What is the success rate of the adapter compared with PoE and CO3 over all held-out pairs and multiple seeds? U-Q2

Reviews.pdf, page 3 (no text layer; passage located by OCR)

these counts

this claim

THE SUBMITTED WEAKNESS

Figure 4, the only counts

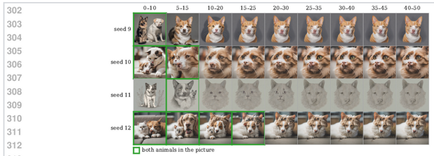


Figure 4: **Composition survives only while the correction window starts early.** A cat and a dog oracle correction, one pair

56173_Restoring_plurality_in_c.pdf (submitted), page 6

Section 4, the held-out claim

pair where joint prompting cannot even produce an image with plurality are left for evaluation of the generalisability of our LoRA module. Neither concept in a held-out pair appears in any training pair. The starting noise used for evaluation is also excluded from training. For every training no concept shared?

56173_Restoring_plurality_in_c.pdf (submitted), page 7

learned adapter, 276 prompts

failure rates

protocol stated

THE REVISED ANSWER

Table 1, Attend-and-Excite prompts

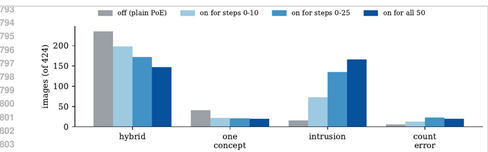
every image scored

Table 1: Composition and image quality on the Attend-and-Excite prompts with SDXL. Rows are methods, all from the same starting noise for each prompt and seed (seeds 9 to 12); column groups are the animals (66 prompts), animals-with-objects (144) and objects (66) prompt sets. Each method has 1,104 images (276 prompts, 4 seeds each). Comp. is the share of animal-prompt images in which a detector counts two or more animals; CLIP is CLIP score (0 to 100), BLIP is BLIP-VQA (0 to 1) and IR is ImageReward; higher is better for all four. The joint prompt, CO3 and ToMe evaluate the joint prompt at every step and sit above the rule as references; bold marks the best value in each column, and ↑ that higher is better. ImageReward is a learned preference score centred near zero, so a negative value means a picture people would rank below average.

Method	Animals				Animals + objects			Objects		
	Comp. ↑	CLIP ↑	BLIP ↑	IR ↑	CLIP ↑	BLIP ↑	IR ↑	CLIP ↑	BLIP ↑	IR ↑
Joint prompt	0.89	27.0	0.71	0.89	**30.0**	**0.88**	1.60	31.1	0.48	0.77
CO3	0.94	27.4	0.74	1.27	29.9	0.87	**1.69**	**31.5**	0.42	0.95
ToMe	0.96	**27.7**	**0.81**	**1.32**	29.6	0.88	1.63	30.4	**0.70**	**1.29**
PoE	0.02	21.6	0.23	−1.31	27.2	0.55	−0.25	29.5	0.45	−0.22
Ours	**0.97**	24.6	0.50	0.16	27.1	0.68	0.42	28.7	0.55	0.19

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 9

Figure 11, failure types counted



failures counted

Figure 11: Failure types as the adapter acts on fewer steps. Each image is labelled by Claude

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 15

Section 4.1, the protocol

fixed before scoring

We evaluate one adapter, chosen once by a fixed rule before any held-out score was read.

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 8

The reply we plan to send (replies/k4nd-and-uhz6.md: k4nd W3 and Q2, UHz6 Q2)

answered

W3: no systematic quantitative evaluation of the adapter

We agree, and the revision adds this evaluation. The adapter, the seeds and the success bar were all fixed before any held-out score was read (Section 4.1).

Table 1 scores every method on the 276 Attend-and-Excite prompts (66 animal pairs, 144 animal-object pairs, 66 object pairs) at seeds 9 to 12, with composition measured by BLIP-VQA and image quality by CLIP score and ImageReward.

The training and held-out pairs, with every seed, are released with the code [TODO: anonymous link].

Hybrids, missing concepts and additional objects are counted per setting in Figure 11 (Appendix C.2).

Q2: aggregate results over held-out concepts and seeds

Answered under k4nd W3: Table 1, and Figure 11, which counts the failure types the reviewer names.

Q2: success rate against PoE and CO3 over all held-out pairs and seeds

Answered under k4nd W3: Table 1, and Figures 11 and 12 for the held-out pairs at eight seeds.

Changes behind this page: Table 1 scores every method on the 276 Attend-and-Excite prompts, Figure 11 counts failure types on 53 held-out pairs, and Section 4.1 opens with the protocol.

UHz6 weakness 1 also asks for standard benchmarks; that half is on the G4 page. The reply's anonymous code link is still a TODO. Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.

G4: Standard benchmarks and metrics

PrzE weakness 1 and question 1, UHz6 weakness 1 · status: pending (Table 2 is rendering)

THE REVIEWER'S POINT

PrzE, weakness 1

- Lack of quantitative benchmarks: The paper does not report a single quantitative metric. Standard benchmarks such as T2I-Compench, GenEval and standard metrics such as CLIP, Imagereward are completely absent in this paper. The empirical results and a few qualitative results are insufficient for the claim in this paper.

Reviews.pdf, page 2 (no text layer; passage located by OCR)

P-W1

PrzE, question 1

1. Quantitative Benchmark: what are the evaluation scores for the proposed method? Can you show the results on some benchmarks and compare your method with other standard baselines?

Reviews.pdf, page 2 (no text layer; passage located by OCR)

P-Q1

UHz6, weakness 1

method as it is difficult to evaluate its generalization. In addition to the examples being limited to similar entities (cat, dog), there are no standardized benchmark evaluations such as T2I-CompBench or GenEval counting.

Reviews.pdf, page 3 (no text layer; passage located by OCR)

U-W1b

no metric

THE SUBMITTED WEAKNESS

Section 4.1, the results

374 Figure 5 shows the results of our method compared to joint prompting, PoE and CO3 (which still
375 uses the joint prompt). Importantly, for inference using our LoRA module the joint prompt is never
376 evaluated, and inference still requires only three predictions per denoising step (one per concept and
377 the null prompt where the LoRA is active within these predictions). As expected, joint prompting

56173_Restoring_plurality_in_c.pdf (submitted), page 7

results are image grids only

scored

benchmarks, pending

THE REVISED ANSWER

Table 1, Attend-and-Excite prompts

CLIP, BLIP-VQA, ImageReward

Table 1: Composition and image quality on the Attend-and-Excite prompts with SDXL. Rows are methods, all from the same starting noise for each prompt and seed (seeds 9 to 12); column groups are the animals (66 prompts), animals-with-objects (144) and objects (66) prompt sets. Each method has 1,104 images (276 prompts, 4 seeds each). Comp. is the share of animal-prompt images in which a detector counts two or more animals; CLIP is CLIP score (0 to 100), BLIP is BLIP-VQA (0 to 1) and IR is ImageReward; higher is better for all four. The joint prompt, CO3 and ToMe evaluate the joint prompt at every step and sit above the rule as references; bold marks the best value in each column, and ↑ that higher is better. ImageReward is a learned preference score centred near zero, so a negative value means a picture people would rank below average.

Method	Animals				Animals + objects			Objects		
	Comp. ↑	CLIP ↑	BLIP ↑	IR ↑	CLIP ↑	BLIP ↑	IR ↑	CLIP ↑	BLIP ↑	IR ↑
Joint prompt	0.89	27.0	0.71	0.89	**30.0**	**0.88**	1.60	31.1	0.48	0.77
CO3	0.94	27.4	0.74	1.27	29.9	0.87	**1.69**	**31.5**	0.42	0.95
ToMe	0.96	**27.7**	**0.81**	**1.32**	29.6	0.88	1.63	30.4	**0.70**	**1.29**
PoE	0.02	21.6	0.23	−1.31	27.2	0.55	−0.25	29.5	0.45	−0.22
Ours	**0.97**	24.6	0.50	0.16	27.1	0.68	0.42	28.7	0.55	0.19

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 9

Table 2, GenEval and T2I-CompBench

pending: slot R14

Table 2: Standard benchmarks on their own prompts with SDXL. GenEval's two-object and counting tasks and T2I-CompBench's spatial and complex tasks, each scored by its official evaluator; higher is better.

R14: result pending

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 15

The reply we plan to send (replies/prze.md: W1 and Q1)

pending: Table 2

W1: lack of quantitative benchmarks

The revision adds quantitative evaluation. Table 1 scores every method on the 276 Attend-and-Excite prompts with BLIP-VQA (the T2I-CompBench composition metric), CLIP score and ImageReward. Table 2 reports GenEval's two-object and counting tasks and T2I-CompBench's spatial and complex tasks with their official evaluators [TODO: slot R14].

Q1: quantitative benchmark

Answered under W1 and W2: Tables 1 and 2.

Table 2 (slot R14) is rendering on the RunPod pod; the page shows its placeholder until it lands. No number is estimated.

The PrzE Q1 reply says "Tables 1 to 4"; the revision has three tables. UHz6's reply to weakness 1 is on the G3 page. Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.

G5: Baselines beyond CO3

PrzE weakness 2, UHz6 weakness 3 · status: partly answered (no number for the Langevin corrector)

THE REVIEWER'S POINT

PrzE, weakness 2

- Narrow baselines and Missing details: There is only CO3 baseline compared, the paper doesn't provide other corrected methods such as MCMC corrector. The paper only compares the baseline CO3 without the corrector they mention. The

P-W2a

Reviews.pdf, page 2 (no text layer; passage located by OCR)

UHz6, weakness 3

- There are no quantitative comparisons with the compositional samplers discussed in Sec. 2.2, and CO3 appears only as one qualitative column. Attention-based methods that address missing or merged objects under joint prompts (e.g., Attend-and-Excite, CONFORM) target the same failure mode and are not discussed. The study is also limited to SDXL

U-W3a

U-W3b

Reviews.pdf, page 3 (no text layer; passage located by OCR)

only CO3

THE SUBMITTED WEAKNESS

Section 4.1, the results

374 Figure 5 shows the results of our method compared to joint prompting, PoE and CO3 (which still
375 uses the joint prompt). Importantly, for inference using our LoKA module the joint prompt is never

CO3 the only baseline

56173_Restoring_plurality_in_c.pdf (submitted), page 7

more columns

discussed

no number

THE REVISED ANSWER

Figure 5, six columns



Figure 5: The adapter repairs the composition without the joint prompt. Rows are four unseen concept pairs, and columns are the joint prompt, the plain product, CO3, ToMe, the product with a Langevin corrector, and our adapter. ToMe keeps the sampler of its release; every other column uses DDIM, and every column starts from the same latent.

ToMe, Langevin added

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 8

Section 2, attention-based methods

213 Attention-based methods such as Attend-and-Excite (Chefer et al., 2023) and CONFORM
214 (Meral et al., 2024) target missing or merged objects under a single joint prompt by adjust-
215 ing cross-attention during sampling; in our setting the joint prompt is never evaluated at
inference, only the separate concept prompts.

now discussed

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 4

Not in the revision

The Langevin corrector appears in Figure 5 only, with no number in a table.

no crop: this panel states what the claim table records

The reply we plan to send (replies/prze.md: W2; replies/k4nd-and-uhz6.md: UHz6 W3)

partly answered

W2: narrow baselines and missing details

Figure 5 now includes a product of experts with a Langevin corrector (Du et al., 2023), the MCMC correction the paper discusses, run at its published setting from the same starting latent, beside CO3, ToMe and our adapter. Table 1 compares with CO3 on every image. Table 1 adds ToMe, run on SDXL from the same starting latents [TODO: its row of Table 1]. Section 4 now describes the training set and how pairs were chosen (Figure 7); the 48 pairs are listed in our reply to Reviewer UHz6 and released with the code.

UHz6's weakness 3 reply (replies/k4nd-and-uhz6.md) says the same: CO3 in Tables 1 and 3, a Langevin corrector in Figure 5, Attend-and-Excite and CONFORM in Section 2.

Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.

G6: The training set and the CO3 filter

PrzE weakness 2, UHz6 weakness 2 and question 1, k4nd question 3 · status: answered

THE REVIEWER'S POINT

UHz6, weakness 2

- For reproducibility concerns, the training set used for adapter training is not provided in sufficient detail. The authors specify the prompts used for training as the concepts that can be generated by Stable Diffusion XL and where joint prompting fails in Sec. 4. In addition, the training filter keeps a pair if either the joint prompt or CO3 succeeds, but the target is always the joint prompt (Eq. 11). Pairs that pass only through CO3 would therefore supervise toward a target without plurality.

U-W2a · U-W2b

Reviews.pdf, page 3 (no text layer; passage located by OCR)

UHz6, question 1

- Could the authors list the 48 training pairs and the held-out pairs, and clarify whether cat/dog is in the training set?

U-Q1

Reviews.pdf, page 3 (no text layer; passage located by OCR)

k4nd, question 3

- Training pairs are retained when either joint prompting or CO3 succeeds, but supervision comes from the frozen joint-prompt prediction. How are pairs handled when only CO3 succeeds?

K-Q3

Reviews.pdf, page 5 (no text layer; passage located by OCR)

PrzE, weakness 2

methods such as MCMC corrector. The paper only compares the baseline CO3 without the corrector they mention. The 48 concept pairs are not listed so it's impossible to see what the adapter is trained on.

P-W2b

Reviews.pdf, page 2 (no text layer; passage located by OCR)

this filter

not listed

THE SUBMITTED WEAKNESS

Section 4, the training pool

More formally, we construct the training pool from concept pairs that Stable Diffusion XL can compose under a joint prompt. A pair that fails under joint prompting cannot provide the desired target. For each candidate pair, we generate one image using plain joint prompting and one using CO3 (Dutta et al., 2026), and retain the pair if either image contains both concepts. This filtering leaves 48 concept pairs which demonstrate plurality from joint prompting. Remaining concept pair where joint prompting cannot even produce an image with plurality are left for evaluation of the generalisability of our LoRA module. Neither

either image passes

48 pairs, none named

56173_Restoring_plurality_in_c.pdf (submitted), page 7

described how

shown

THE REVISED ANSWER

Section 4, the training set

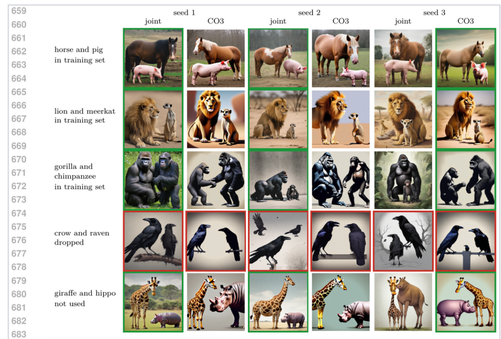
every target is the joint prompt

CO3 a second view only

48 pairs (28 of two animals, 3 of an animal and an object, 17 of two objects) at 182 pairs-and-seed cells over 23 seeds, and every target is the frozen model's joint-prompt prediction. Pairs were kept when the joint prompt drew both named things as separate objects, and left out when the two concepts could not be told apart (Figure 7; the full list is released with the code). Candidates were screened with the joint prompt and, for some, with CO3 (Dutta et al., 2026) as a second view; no CO3 picture is a training target. No pair in Figure 5 was trained on. Seeds 9 to 16 are never trained on. For every training pair and seed, we

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 7

Figure 7, screening with CO3 beside



joint prompt picks the pair

Figure 7: Screening candidate pairs with the joint prompt and CO3. Each seed shows the

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 13

The reply we plan to send (replies/k4nd-and-uhz6.md: UHz6 training filter and Q1, k4nd Q3)

answered

The training filter

We have addressed this in Section 4 and Figure 7. Every training target is the frozen model's joint-prompt prediction; no CO3 picture is a target. CO3 was a second view during screening: a pair entered the training set when the joint prompt drew both named things as separate objects. Figure 7 shows the joint prompt beside CO3 for five candidate pairs, including crow and raven, dropped because the two birds are hard to tell apart. Where CO3 drew a kept pair more clearly on a seed (seed 3 in Figure 7), that seed's target is still the joint prompt's picture.

Q1: the 48 training pairs, the held-out pairs, and whether cat and dog is trained on

The training set is defined by a criterion, not by the particular pairs. A pair is kept when the frozen model's joint prompt draws both named concepts as separate objects, and dropped when the two concepts cannot be told apart (Section 4, Figure 7). The same criterion applies whether it yields 48 pairs or a thousand, and 48 is where we stopped collecting. The method relies on the criterion, not on any specific pair.

Cat and dog is not in the training set. It is held out, and no training pair contains a cat or a dog. The held-out pairs are released with the code, and no pair in Figure 5 was trained on.

For reproducibility, the 48 training pairs are: (two animals, 28) badger and magpie, bear and boar, bear and heron, beaver and raccoon, beaver and woodpecker, bison and weasel, buffalo and hippo, bull and rooster, camel and pelican, chicken and rabbit, deer and heron, deer and wolf, donkey and duck, donkey and rooster, emu and koala, giraffe and hyena, giraffe and ostrich, giraffe and zebra, gorilla and chimpanzee, hedgehog and robin, horse and pig, horse and sheep, impala and jackal, langur and peacock, lion and meerkat, lion and oryx, moose and chipmunk, peacock and pig; (an animal and an object, 3) hamster and carrot, pony and red barn, snail and lantern; (two objects, 17) accordion and scarecrow, bicycle and birdhouse, birdcage and hay bale, cactus and cooler box, coat rack and potted plant, guitar and traffic cone, hay bale and ukulele, lighthouse and sailboat, mailbox and pumpkin, microscope and test tube, rocking chair and cactus, soccer ball and park bench, suitcase and globe, teddy bear and birthday cake, telescope and beach ball, typewriter and cactus, windmill and tractor. They are also released with the code as the configuration file the trainer reads.

Q3: pairs retained when only CO3 succeeds

We have addressed this in Section 4 and Figure 7. Every training target is the frozen model's joint-prompt prediction; no CO3 picture is a target. CO3 was a second view during screening: a pair entered the training set when the joint prompt drew both named things as separate objects. Figure 7 shows the joint prompt beside CO3 for five candidate pairs, including crow and raven, dropped because the two birds are hard to tell apart. Where CO3 drew a kept pair more clearly on a seed (seed 3 in Figure 7), that seed's target is still the joint prompt's picture.

The 48 pairs are listed in the UHz6 Q1 reply and released with the code; the paper describes them by kind (28 two animals, 3 animal and object, 17 two objects).

Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.

G7: The training loss and what the adapter returns

k4nd weakness 4 (two parts) and question 4 · status: answered

THE REVIEWER'S POINT

k4nd, weakness 4

4. Essential method details are poorly presented and sometimes inconsistent.

Section 4 describes target matching and provides the teacher prediction and sampling rule, but it does not explicitly formulate the training loss. The squared-error objective appears in the training diagram in Appendix A, Figure 8. Essential information about the optimization should be stated clearly in the main method section.

There are also substantive ambiguities. The main text describes individual adapted network evaluations as returning the correction, whereas Appendix B defines them as adapted denoising predictions whose composition induces the correction. Section 4 refers to "Figure 10," although the actual training diagram is Figure 8.

Reviews.pdf, page 5 (no text layer; passage located by OCR)

K-W4a · K-W4a · K-W4b

k4nd, question 4

- Can the authors state the optimization objective explicitly in the main text and resolve the inconsistencies concerning network outputs, figure references, and baseline starting noise?

Reviews.pdf, page 5 (no text layer; passage located by OCR)

K-Q4

this sentence · no loss here · the diagram

THE SUBMITTED WEAKNESS

Section 4, opening paragraph

We train a LoRA module (Hu et al., 2022) on the query, key, and value projections of every cross-attention of layer of Stable Diffusion XL (Podell et al., 2024) while keeping the model's existing weights frozen. This LoRA module is denoted by $\epsilon_{\theta,\psi}$ with ψ as the LoRA parameters. In these layers, queries are computed from the latent, while keys and values are computed from the prompt tokens. The module is trained to output the plurality correction for a given latent representation. If the module successfully learns the abstract notion of plurality, then it can be included during inference time with the usual PoE denoising step such that a joint prompt is not needed. During training, the joint prompt provides the target. Each network evaluation takes a latent x_t, its timestep t, and one prompt, and returns the predicted correction defined in Equation 9. The adapted composition $\hat{\epsilon}^{\psi}_{\text{PoE}}$ uses separate predictions under c_1, c_2, and the null prompt. It combines them using the same guided product-of-experts rule as the unadapted model with the estimated correction term as shown in Figure 10.

56173_Restoring_plurality_in_c.pdf (submitted), page 7

returns the correction

Section 4, how the adapter is trained

Figure 5: **The adapter repairs the composition without the joint prompt.** Rows are four unseen concept pairs, and columns are the joint prompt, plain product, CO3, and our adapter. CO3 uses different starting noise; the other three columns share it.

to the frozen model's guided prediction under the joint prompt c_{12}:

$$\hat{\epsilon}_{\text{Mono}} = \epsilon_\theta(x_t, t \mid \emptyset) + w\big[\epsilon_\theta(x_t, t \mid c_{12}) - \epsilon_\theta(x_t, t \mid \emptyset)\big]. \tag{11}$$

Only the adapter parameters ψ are updated. At the selected states, the target differs from the unadapted composition by r_t, so matching it trains the adapter to supply that difference. During sampling, the learned correction is the change that the adapter makes to the composed prediction. We denote it by $\hat{r}_\psi = \hat{\epsilon}^{\psi}_{\text{PoE}} - \hat{\epsilon}_{\text{PoE}}$ and apply it in the same form as the measured correction:

$$\hat{\epsilon}_\lambda = \hat{\epsilon}_{\text{PoE}} + \lambda \hat{r}_\psi = \hat{\epsilon}_{\text{PoE}} + \lambda\left(\hat{\epsilon}^{\psi}_{\text{PoE}} - \hat{\epsilon}_{\text{PoE}}\right). \tag{12}$$

Appendix A gives the training and sampling details. Appendix B derives the relation between the training target and the prediction used during sampling.

56173_Restoring_plurality_in_c.pdf (submitted), page 7

no loss written · applied as a correction at strength λ

Appendix A, Figure 8

Figure 8 shows how these runs are trained. Frozen SDXL first supplies the joint-prompt target on states cached from a plain product-of-experts trajectory. At each selected state, the adapted network evaluates the two concept prompts and the null token separately; their composed prediction is compared with that target. The loss updates only the LoRA weights. The diagram separates the one-time target cache from the three adapted predictions and the loss used to update the LoRA weights.

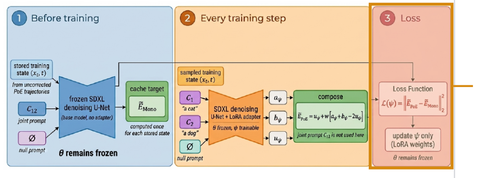


Figure 8: **Training the compositional adapter.** A cached frozen joint-prompt target supervises the adapted product-of-experts prediction; only the LoRA weights are updated. The frozen network supplies the cached target; the adapted network produces the composition matched to it.

56173_Restoring_plurality_in_c.pdf (submitted), page 12

the squared error

now reads · replaced by · Equation 12 added · stated in Section 4 instead

THE REVISED ANSWER

Section 4, opening paragraph

layers, all three are computed from the latent. Once trained, the module is used at inference with the usual PoE step, so a joint prompt is not needed. During training, the joint prompt provides the target. Each network evaluation takes a latent x_t, its timestep t and one prompt, and returns a noise prediction, as the frozen model does. At inference we run the usual product-of-experts step on c_1, c_2 and the null prompt with the adapter switched on (Equation 14): three predictions per step, and never the joint prompt.

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 7

returns a noise prediction · inference: PoE, adapter on

Section 4, Equation 12

Only the adapter parameters ψ are trained, by minimising

$$\mathcal{L}(\psi) = \mathbb{E}_{x_t,t}\left\| \hat{\epsilon}^{\psi}_{\text{PoE}}(x_t, t) - \hat{\epsilon}_{\text{Mono}}(x_t, t) \right\|^2 \tag{12}$$

over the cached states of every training pair and seed at the first ten of the 50 sampling steps. The joint prompt's prediction is the product's prediction plus r_t. Equation 12 therefore trains the adapter to add r_t to the product. At inference, switching the adapter on adds it. Appendix A gives the training and sampling details. Appendix B derives the

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 8

Equation 12, the training loss · trains the adapter to add r_t

---

The reply we plan to send (replies/k4nd-and-uhz6.md: k4nd W4, its first two parts, and Q4)

answered

W4: method details poorly presented and sometimes inconsistent: The training loss is not stated in Section 4

Section 4 now states the training loss as Equation 12: the squared error between the adapted composition and the joint-prompt prediction, over the cached states of every training pair and seed at the first ten of the 50 sampling steps. The training diagram is removed, since Equation 12 now states the objective in the main text.

W4: method details poorly presented and sometimes inconsistent: Evaluations described as returning the correction

Section 4 now says each adapted evaluation returns a noise prediction, as Appendix B derives. At inference we run the usual product-of-experts step on c_1, c_2 and the null prompt with the adapter switched on: three predictions per step, and never the joint prompt. The joint prompt's prediction is the product's prediction plus the correction r_t, so matching it trains the adapter to add r_t. At inference, switching the adapter on adds it. The reference that printed as "Figure 10" is removed.

Q4: the objective in the main text, and the inconsistencies

The objective is Equation 12 in Section 4. The network outputs and the figure reference are resolved as described under W4. Baseline starting noise is resolved as described under W5.

Changes behind this page: 8ef00a17 Section 4 states the training loss as Equation 12 · 88f02a45 and ead813ab each adapted evaluation returns a noise prediction and inference is the usual PoE step with the adapter on · 2e9af75c the training diagram is hidden.
Question 4 also asks about figure references and starting noise; those are shown on the G9 and G8 pages. The reviewer's quotations are left out of the reply panel because the left column shows them.

G8: The same starting noise for every method

k4nd weakness 5 · status: answered

THE REVIEWER'S POINT

k4nd, weakness 5

5. The qualitative baseline comparison contains a protocol inconsistency.

The Figure 5 caption states that CO3 uses different starting noise, while Section 4.1 states that all examples use the same initial seeds. This discrepancy matters because the text attributes changes in visual style to CO3. With different initial noise, the displayed comparison cannot isolate that effect. A clearly specified comparison over multiple seeds is necessary.

Reviews.pdf, page 5 (no text layer; passage located by OCR)

K-W5

K-W5

this caption

this sentence

THE SUBMITTED WEAKNESS

Figure 5 caption

356 Figure 5: **The adapter repairs the composition without the joint prompt.** Rows are four
357 unseen concept pairs, and columns are the joint
358 prompt, plain product, CO3, and our adapter.
359 CO3 uses different starting noise; the other three
360 columns share it.
361

CO3 different noise

56173_Restoring_plurality_in_c.pdf (submitted), page 7

Section 4.1

381 uncommon pairing. Importantly, all examples are generated from the same initial seeds. The second
382 column shows the corresponding product-of-experts samples. In each case, the two concepts are not

same seeds

56173_Restoring_plurality_in_c.pdf (submitted), page 8

now

now

THE REVISED ANSWER

Figure 5 caption

same latent, every column

401 Figure 5: The adapter repairs the composition without the joint prompt. Rows are four
402 unseen concept pairs, and columns are the joint prompt, the plain product, CO3, ToMe, the
403 product with a Langevin corrector, and our adapter. ToMe keeps the sampler of its release;
404 every other column uses DDIM, and every column starts from the same latent.

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 8

Section 4.1, the protocol

CO3 from the same latent

421 We evaluate one adapter, chosen once by a fixed rule before any held-out score was read,
422 against joint prompting, PoE, CO3 and ToMe. Every method uses the same seed for each
423 prompt, with 50 steps at guidance 7.5 (DDIM, except ToMe, which keeps the sampler of its
424 release); CO3 starts from the same latent but draws its later resampling noise from the seed,

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 8

Section 4.1, continued

CO3 column marked †

424 release); CO3 starts from the same latent but draws its later resampling noise from the seed,
425 so its column is marked †. Table 1 scores the Attend-and-Excite prompts (276 prompts, seeds

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 8

The reply we plan to send (replies/k4nd-and-uhz6.md: k4nd W5)

answered

W5: the qualitative baseline comparison and its starting noise

Every method now uses the same seed for each prompt, with every column of a row in Figure 5 starting from the same latent, and Section 4.1 says so. CO3's column is marked † because its later resampling noise comes from the seed, as in its released code. Table 1 makes the comparison over 4 seeds, and Figures 10 and 11 over 8, with the images behind them in the appendix.

k4nd's question 4 also asks about starting noise; it is answered here. Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.

G9: A missing figure, a broken sentence, repeated claims

UHz6 weakness 4, k4nd weakness 4 (last two parts) · status: answered

THE REVIEWER'S POINT

UHz6, weakness 4

- There are presentation issues: Sec. 4 refers to a Figure 10 that does not exist, and several sentences are broken (e.g., "inside create hybrid objects" in Sec. 5).

Reviews.pdf, page 3 (no text layer; passage located by OCR)

k4nd, weakness 4 (continued)

Section 4 refers to "Figure 10," although the actual training diagram is Figure 8.

The repetitive claims about learning an abstract concept occupy space that could instead clarify the objective, training data, evaluation protocol, and limitations. These presentation problems make the method unnecessarily difficult to assess.

Reviews.pdf, page 5 (no text layer; passage located by OCR)

U-W4a · U-W4b · K-W4c · K-W4d

this reference · repeated claim · this sentence

THE SUBMITTED WEAKNESS

Section 4, opening paragraph

330 the prompt tokens. The module is trained to output the plurality correction for a given latent
331 representation. If the module successfully learns the abstract notion of plurality, then it can be
332 included during inference time with the usual PoE denoising step such that a joint prompt is not
333 needed. During training, the joint prompt provides the target. Each network evaluation takes a latent
334 x_t, its timestep t, and one prompt, and returns the predicted correction defined in Equation 9. The
335 adapted composition $\tilde{\epsilon}^{\emptyset}_{\text{PoE}}$ uses separate predictions under c_1, c_2, and the null prompt. It combines
336 them using the same guided product-of-experts rule as the unadapted model with the estimated
337 correction term as shown in Figure 10.

abstract notion of plurality · Figure 10 does not exist

56173_Restoring_plurality_in_c.pdf (submitted), page 7

Discussion

463 In this work we introduce the concept of plurality: the notion that distinct objects in a prompts should
464 become distinct objects in a generated image. We show that PoE and similar compositional diffusion
465 methods are unable to create images with plurality, inside create hybrid objects that combine the in-
466 dividual objects being prompted for. We show that plurality can be restored during early denoising

broken sentence

56173_Restoring_plurality_in_c.pdf (submitted), page 9

both removed · now reads

THE REVISED ANSWER

Section 4, opening paragraph

no Figure 10, no abstract notion

560 layers, all three are computed from the latent. Once trained, the module is used at
561 inference with the usual PoE step, so a joint prompt is not needed. During training, the
562 joint prompt provides the target. Each network evaluation takes a latent x_t, its timestep
563 t and one prompt, and returns a noise prediction, as the frozen model does. At inference
564 we run the usual product-of-experts step on c_1, c_2 and the null prompt with the adapter
switched on (Equation 14): three predictions per step, and never the joint prompt.

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 7

Discussion

sentence repaired

486 similar compositional diffusion methods are unable to create images with plurality, and
487 instead create hybrid objects that combine the individual objects being prompted for. We

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 10

The reply we plan to send (replies/k4nd-and-uhz6.md: UHz6 presentation issues, k4nd W4 third part)

answered

Presentation issues

The reference that printed as "Figure 10" in Section 4 is removed, and the Discussion now reads "and instead create hybrid objects".

W4: method details poorly presented and sometimes inconsistent: Repetitive claims about learning an abstract concept

We removed the repeated claims about learning an abstract concept from Sections 3 and 4 and the Discussion. The introduction now states what the correction is and what it restores (see W1). Section 4 now describes the training data and how pairs were chosen (Figure 7). Appendix C.5 (Figure 15) and a new Discussion paragraph show the adapter on three concepts, including where it fails: it separates the concepts in every case but stops at two animals. Section 4.1 now opens with the evaluation protocol: one adapter chosen by a fixed rule, the same seed for every method, the prompt sets and held-out pairs with their seeds, the scorers, and the renders behind Table 1 in the appendix.

The repeated abstract-concept claims were cut from Sections 3 and 4 and the Discussion (06a9cf98); Section 4.1 opens with the evaluation protocol (508d2647). Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.

G10: Beyond two animals on SDXL

PrzE question 2, UHz6 weakness 3 (scope) · status: partly answered (SDXL only)

THE REVIEWER'S POINT

PrzE, question 2

2. Multiple concepts: How does your method perform when composing more than two concepts? Can you show some results for more than two concepts?

P-Q2

Reviews.pdf, page 2 (no text layer; passage located by OCR)

UHz6, weakness 3 (scope)

Attend-and-Excite, CONFORM) target the same failure mode and are not discussed. The study is also limited to SDXL and to pairs of two concepts, mostly animals.

U-W3c

Reviews.pdf, page 3 (no text layer; passage located by OCR)

pairs only

THE SUBMITTED WEAKNESS

Section 4, the training pool

More formally, we construct the training pool from concept pairs that Stable Diffusion XL can compose under a joint prompt. A pair that fails

pairs on SDXL only

56173_Restoring_plurality_in_c.pdf (submitted), page 7

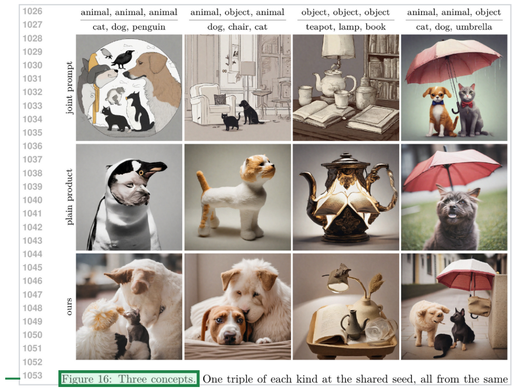
three concepts

discussed

still SDXL

THE REVISED ANSWER

Figure 16, three concepts



one triple of each kind

Figure 16: Three concepts. One triple of each kind at the shared seed, all from the same

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 20

Discussion, three concepts

stops at two animals

The adapter is trained on pairs, yet it carries some of its effect to three concepts (Figure 16). On one triple of each kind, the plain product draws a single fused object, while the adapter separates the concepts in every case and draws all three named things when the triple includes objects. When the triple has two or three animals it stops at two, which suggests that what it learned is closer to "two separate things" than to a count. Whether this holds across seeds, and whether training on triples would close the gap, are open questions.

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 10

Not in the revision

No model other than SDXL. The SD 1.5 results were removed on 2026-11-12.

no crop: this panel states what the claim table records

The reply we plan to send (replies/prze.md: Q2)

partly answered

Q2: multiple concepts

Appendix C.4 (Figure 16) shows one triple of each kind. The plain product fuses the three concepts or drops one; the adapter, trained only on pairs, separates them in every case and draws all three named things when the triple includes objects, but with two or three animals it draws only two. The Discussion now addresses this.

Objects beyond animals are scored in Table 1 (animals with objects, objects); see G4. UHz6's weakness 3 reply is on the G5 page. Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.

G11: The adapter on the early steps only

UHz6 question 3 · status: answered

THE REVIEWER'S POINT

UHz6, question 3

- Given the finding in Sec. 3, does restricting the adapter to the early steps change the failure modes in Fig. 7?

U-Q3

Reviews.pdf, page 3 (no text layer; passage located by OCR)

these failures

THE SUBMITTED WEAKNESS

Figure 7, limitations



Figure 7: **Limitations of the Plurality LoRA:** a) The adapter can change the render under joint prompting and is inconsistently useful in this case. Each row compares the model conditioned on the joint prompt with and without the LoRA applied from the same noise sample. While the LoRA improves the render for a chessboard and hourglass, it disrupts the render for a cat and a dog, and a tiger and a dog. b) We show five examples of how a corrected cat–dog render can lead to inconsistent or unexpected results. The uncorrected hybrid (top left) is followed by five corrected failures showing intrusions (top middle and bottom right), counting errors (bottom middle), style-dependent loss of plurality (bottom left) or instability (top right).

56173_Restoring_plurality_in_c.pdf (submitted), page 9

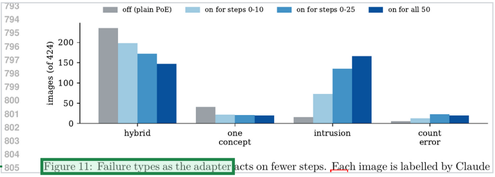
failures, no count

counted per window

shown per window

THE REVISED ANSWER

Figure 11, failure types by window

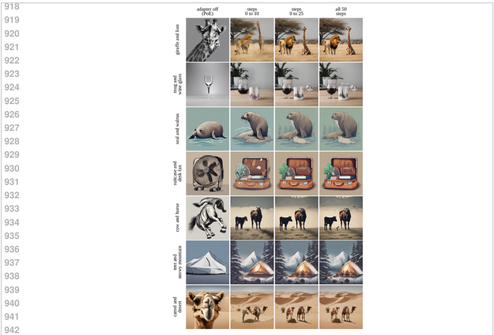


steps 0 to 10, 0 to 25, all 50

Figure 11: Failure types as the adapter acts on fewer steps. Each image is labelled by Claude

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 15

Figure 13, the images behind it



the renders

Figure 13: The images behind Figure 11. Seven held-out pairs, each at the seed where

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 18

The reply we plan to send (replies/k4nd-and-uhz6.md: UHz6 Q3)

answered

Q3: restricting the adapter to the early steps

We tested this directly. Figure 11 counts each failure type on the 53 held-out pairs at seeds 9 to 16 with the adapter switched on for the first 10, the first 25 or all 50 of the 50 steps, and Figure 13 shows the images. The first 25 steps give the same result as all 50. With the first 10 only, the detector finds two animals far less often (Figure 12a). The cow and horse row of Figure 13 shows that the species substitution of the submitted Fig. 7 (Figure 15 in the revision) persists at every window, so it does not come from the late steps.

The submitted Figure 7 is Figure 15 in the revision. Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.

G12: Pairs that do not compete for one region

UHz6 question 4 · status: answered in the reply

THE REVIEWER'S POINT

UHz6, question 4

- Does the adapter degrade PoE on pairs that do not compete for the same region, such as foreground/background pairs like lighthouse and sea?

Reviews.pdf, page 3 (no text layer; passage located by OCR)

U-Q4 → one image

THE SUBMITTED WEAKNESS

Section 4.1, one lighthouse

383 preserved as distinct entities: the animals fuse, the tent and mountain collapse into one cone-like
384 shape. The lighthouse and sea appears structurally correct but the lighthouse itself is reduced to a
385 distant object in the background. This reproduces the results of Bradley et al. (2025) which show
386 that PoE will combine objects when they compete for the same position in an image and works
387 reasonably well when there is a distinction between which objects should fall into the foreground

56173_Restoring_plurality_in_c.pdf (submitted), page 8

one example, no count → answered

THE REVISED ANSWER

Answered in the reply

The revision has no per-group table. The reply answers from the 21 held-out pairs whose concepts do not compete for one region.

no crop: this panel states what the claim table records

The reply we plan to send (replies/k4nd-and-uhz6.md: UHz6 Q4)

answered in the reply

Q4: pairs that do not compete for the same region

No. Among our held-out pairs, 21 have concepts that do not contend for one region, including lighthouse and a stormy sea. On these pairs the adapter scores the same as the plain product within the 95% intervals (BLIP-VQA, seeds 9 to 16), so on pairs the product already composes it neither helps nor harms.

The held-out groups were removed from the paper; the hard, easy and not-competing split is named as future work in the Discussion. Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.